



Society of Petroleum Engineers

SPE-229897-MS

Water Injection Conformance Control using Nitrogen Foam – A Successful Field Trial in a Tight Carbonate Reservoir in Northern Oman

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229897-MS](https://doi.org/10.2118/229897-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Natih reservoir of the D oilfield of block 5 in northern Oman is a tight carbonate reservoir characterized by low permeability of (1-5) md, with a formation temperature of 70°C and high formation water salinity of around 180,000 mg/L. After 15 years of water-flood with horizontal well patterns, field performance review shows typical challenges such as high water-cut and ineffective water injection caused by the water circulation. Due to low permeability and strong reservoir heterogeneity, conventional water conformance methods such as polymer are not applicable. As an alternative, nitrogen foam flooding – a technique using nitrogen and surfactant to create foam to improve water injection conformance – offers a promising solution to control water channeling, reduce water cut and enhance oil recovery.

This technology has two main features: First, nitrogen and surfactant generate a large volume of foam within the reservoir, which temporarily blocks and diverts water flow paths, thereby expanding the sweep efficiency. At the same time, the surfactant reduces the interfacial tension between crude oil and formation brine, enhancing the mobility of crude oil, ultimately improving oil recovery. Second, nitrogen not only replenishes reservoir energy by providing drive, but also tends to accumulate at the top of the reservoir to displace residual oil in the upper sections, hence creating a secondary profile modification.

The formula of foam surfactant was optimized in the lab based on the field crude oil and brine sample. The selected surfactant offers excellent tolerance to high temperatures and high salinity, enabling its application in reservoirs with temperatures up to 100°C and formation water salinity reaching 200,000 mg/L. In September 2024, a total of 355,000 Nm³ of nitrogen and 3,240 m³ of surfactant solution were injected into a horizontal well pattern with one injector and two producers, which resulted in an approximate 10% reduction in the water cut. A comprehensive monitoring program – including well production testing, injection wellhead pressure monitoring and time-lapse production logging – was executed to systematically evaluate well performance. Following foam injection, a clear and sustained increase in injection well pressure

was observed, lasting for approximately three months. Concurrently, memory production logging (MPLT) results indicated an improved and more uniform water injection profile as a result of foam conformance control. By June 2025, the pattern had achieved a cumulative oil gain of 6,630 barrels, meeting the expected incremental oil and UTC targets. This field trial has demonstrated the feasibility and potential of nitrogen foam technology for water injection profile control, providing valuable field data and a solid foundation for its future expansion in such tight carbonate reservoirs.

Introduction

Oil recovery from tight carbonate reservoirs presents formidable challenges, especially in environments characterized by low matrix permeability and complex heterogeneity. The Natih reservoir of the D oilfield in Block 5, Northern Oman, exemplifies these difficulties, with a permeability range of only 1–5 mD, formation temperatures around 70 °C, and high salinity brine reaching 180,000 mg/L. After more than a decade of conventional water flooding using horizontal line-drive well patterns of 50m narrow spacing, the reservoir continues to exhibit performance limitations such as excessive water production and ineffective sweep, primarily due to water channeling and poor conformance, shown in Fig. – 1.

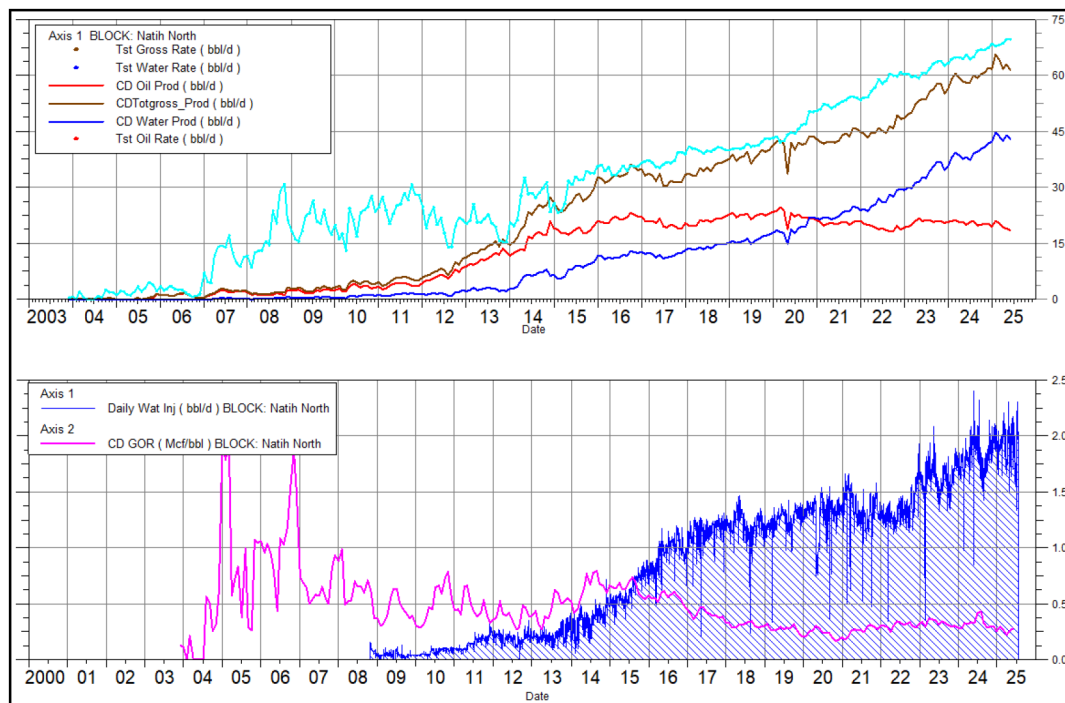


Figure 1—Historical production and injection performance of the studied field

Traditional polymer-based water control methods have proven unsuitable for such extreme conditions, prompting the need for alternative enhanced oil recovery (EOR) strategies. One such promising technique is nitrogen foam flooding, which leverages the mobility control properties of nitrogen-surfactant foam to improve sweep efficiency (Farajzadeh, R., et al., 2009), reduce water cut, and stimulate oil production. This paper explores the implementation and evaluation of nitrogen foam injection in the Natih reservoir, focusing on foam formulation optimization, pattern selection, field-scale performance monitoring, and the effectiveness of the approach in addressing long-standing water flooding inefficiencies in tight, high-salinity and heterogeneous carbonate environments.

This study evaluates a single injector supporting two adjacent, horizontal, line-drive shaped wells in a 50m-spacing pattern. The two producers experience high water cut increase that prompted the need for a solution to rest the WC. Nitrogen foam was injected by surface mixing nitrogen with a foaming

agent, creating a pull-head injection system. A comprehensive surveillance program monitored pressure responses, nitrogen purity, wellhead sampling, and well testing. Temperature log data before and after the trial confirmed effective sweep efficiency. The injection lasted 24 days, followed by a 9-month monitoring period. The trial resulted in a cumulative oil production of 6,630 barrels.

The subsequent sections detail the chemical mechanisms, laboratory validation, and pilot trial design that form the basis of the nitrogen foam flooding approach.

Chemical Working Mechanism

To understand how nitrogen foam functions effectively in the reservoir, it is important to examine its underlying chemical and physical mechanisms in terms of below areas:

Jamin Effect

When foam moving through the micro-pores and micro-fractures in the reservoir, due to the Jamin effect, it will generate very great moving resistance from macro-view, so the foam can block the water channels and inhibit water channeling to increase the sweep efficiency of water flood, (Perm Inc., 2016–2017).

Conformance Control by Foam

Nitrogen foam is a non-Newtonian fluid, and its properties – such as viscosity and stability – are highly sensitive to shear forces. In low-permeability formations, the shear action is more intense, which tends to reduce foam viscosity and stability. Conversely, in high-permeability zones, the shear effect is weaker, allowing the foam to maintain higher viscosity and better stability, (Li, W. et al., 2022), (Zhang, Q. et al., 2023). This behavior of stronger blocking capability of the foam in high-permeability areas compared to low-permeability zones is shown in Fig. – 2.

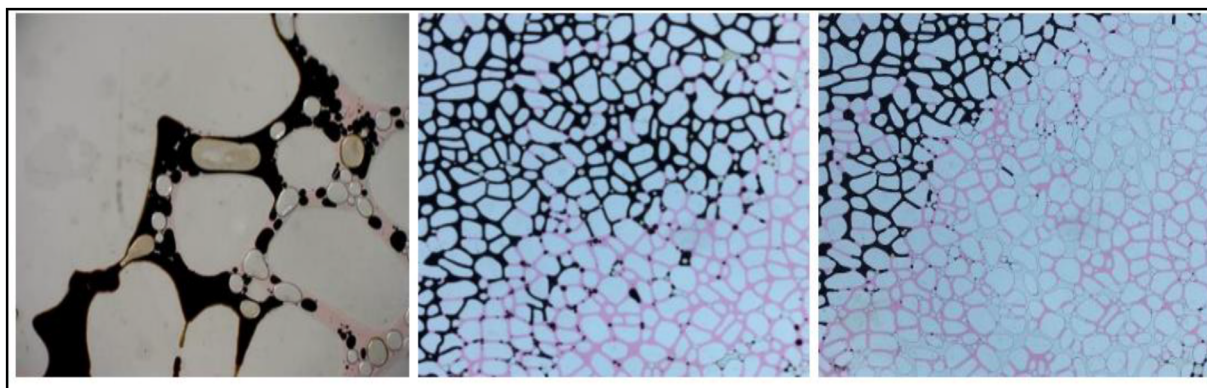


Figure 2—Microfluidic glass etching experiment to illustrate foam conformance control

IFT Reduction

When the foam comes into contact with oil in the reservoir, the surfactants at the gas-liquid interface tend to migrate and adsorb onto the oil-water interface. As a result, the interfacial tension (IFT) between oil and water can be reduced to as low as 10^{-1} dyn/cm, thereby improving the microscopic displacement efficiency. This has indirect effect to the system wettability, which can affect fluid displacement (Lei, W., et al., 2023).

Nitrogen Gas Drive

Compared to water, nitrogen has much higher compressibility, allowing it to expand or compress with pressure changes in the reservoir, which enhances oil displacement efficiency, (Haynes et al., 2008; Alcocer & Menzie, 1984). Additionally, due to its low density, nitrogen released from broken foam bubbles tends to rise to the top of the reservoir, helping to displace residual oil in zones that are difficult for water to reach.

These complementary mechanisms collectively explain why nitrogen foam has the potential to improve sweep efficiency and recovery in tight carbonate reservoirs, motivating laboratory-scale evaluation.

Laboratory Study

To verify these mechanisms, extensive laboratory experiments were carried out, covering blocking ability, foam stability, surfactant selection, gas–liquid ratio optimization, IFT reduction and high temperature & pressure impact. For example, laboratory experiments were conducted to evaluate its blocking ability and stability under various conditions to demonstrate the conformance control mechanism of nitrogen foam. On the other hand, the formulation of the foaming agent must be optimized to suit these challenging conditions such as the high salinity and temperature of the reservoir (Al Yaari, N., & Zitha, P. L. J., 2020). The most effective foaming agent was selected based on its foaming ability and stability when tested with the formation water from the oilfield. Additionally, the interfacial tension (IFT) between oil and water was measured in the laboratory using the crude oil.

Blocking Ability Experiment

In this section, both single-core and dual-core parallel flooding experiments were conducted to evaluate the blocking ability of nitrogen foam. In the single-core test, the inlet and outlet pressures of the core holder were measured and recorded. For the dual-core parallel flooding experiment, one core represented high permeability and the other low permeability, with the flow rate through each core monitored individually, shown in Fig. – 3. The injection scheme is presented in Table 1, while the experimental results are shown in Fig. – 4 and Fig. – 5.

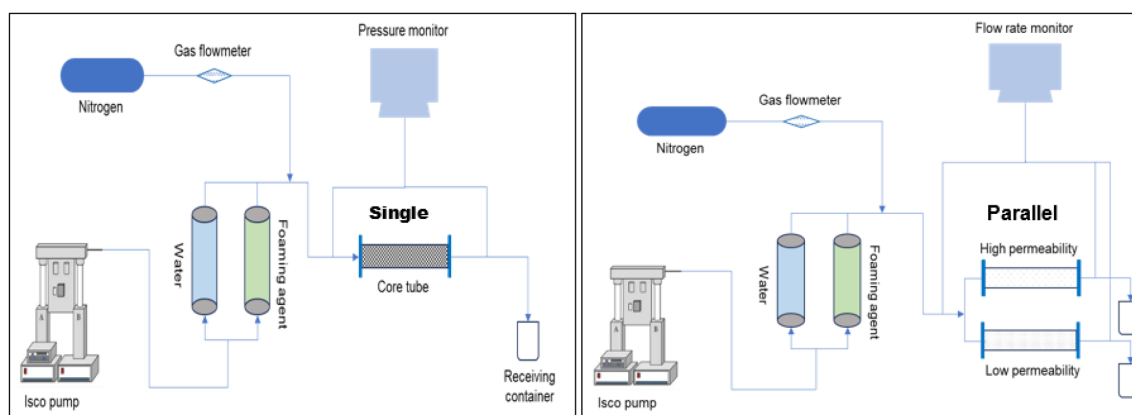


Figure 3—Schematic diagram of single and parallel core flooding experiment

Table 1—Lab experimental injection scheme

Experiment	Injection plan			
	Slug 1	Injection rate	Slug 2	Injection rate
Single core	Water	0.5mL/min	Foam	0.5mL/min
Double cores	Foam	0.5mL/min	Water	1mL/min

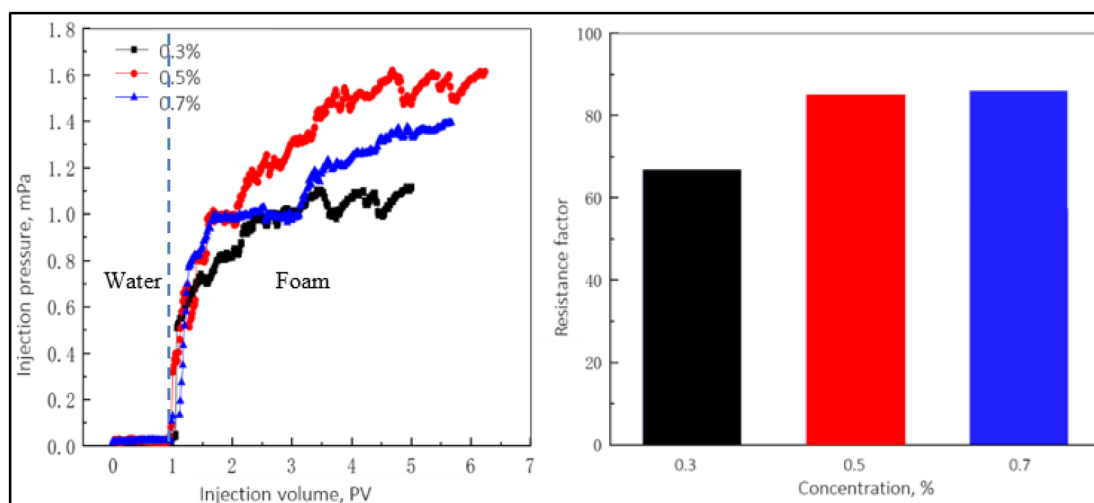


Figure 4—Injection pressure and resistance factor at different concentration

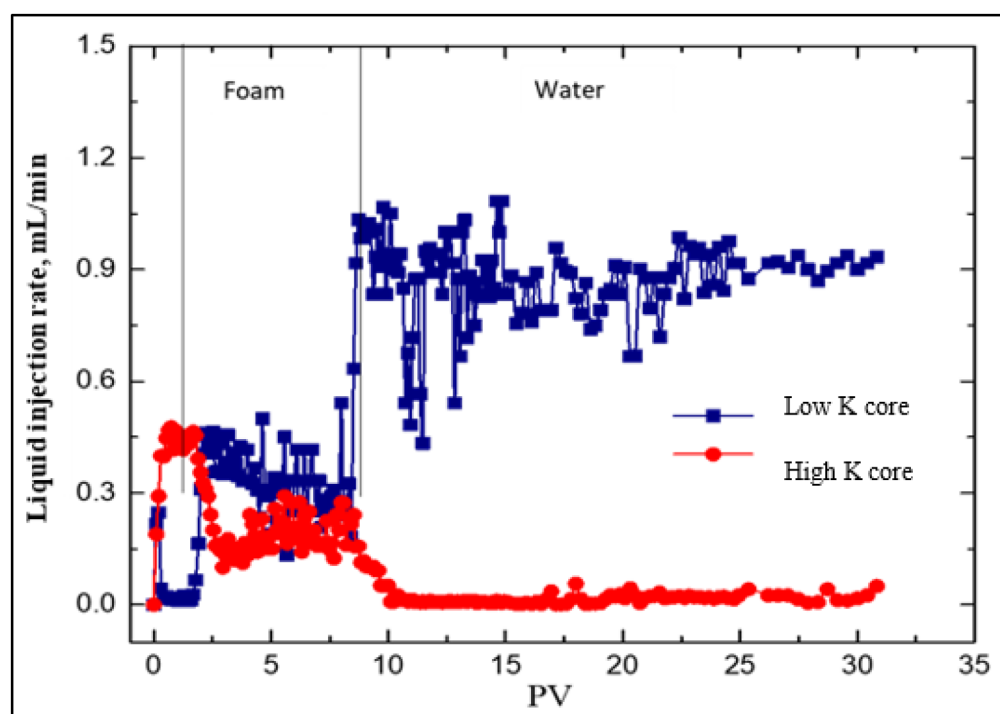


Figure 5—Liquid injection rate after the foam injection

In the single-core flooding experiment, the injection pressure of foam was much higher than that of water at all three foaming agent concentrations. The resistance factors of foam at each concentration exceeded 60, indicating that the foam exhibits excellent blocking ability in porous media.

In the parallel-core flooding experiment, liquid production initially occurred almost entirely from the high-permeability core during the early stage of foam injection. However, after approximately 1 PV of foam injection, the production profile began to shift. Over the subsequent 7 PV of foam injection, the low-permeability core consistently exhibited a higher liquid production rate than the high-permeability core. Upon switching back to water injection, the high-permeability core produced almost no liquid, indicating that the foam provided stronger blocking ability in the high-permeability zone compared to the low-permeability zone.

Foam Stability Experiment

To evaluate the stability of foam in the presence of oil and water, a parallel-core flooding experiment was also conducted, shown as Fig. – 6. In this test, one core was saturated with oil and the other with water. A total of 8 PV of foam followed by 2 PV of water was injected at a constant rate of 3 mL/min. The liquid production rate from each core was measured and recorded throughout the experiment.

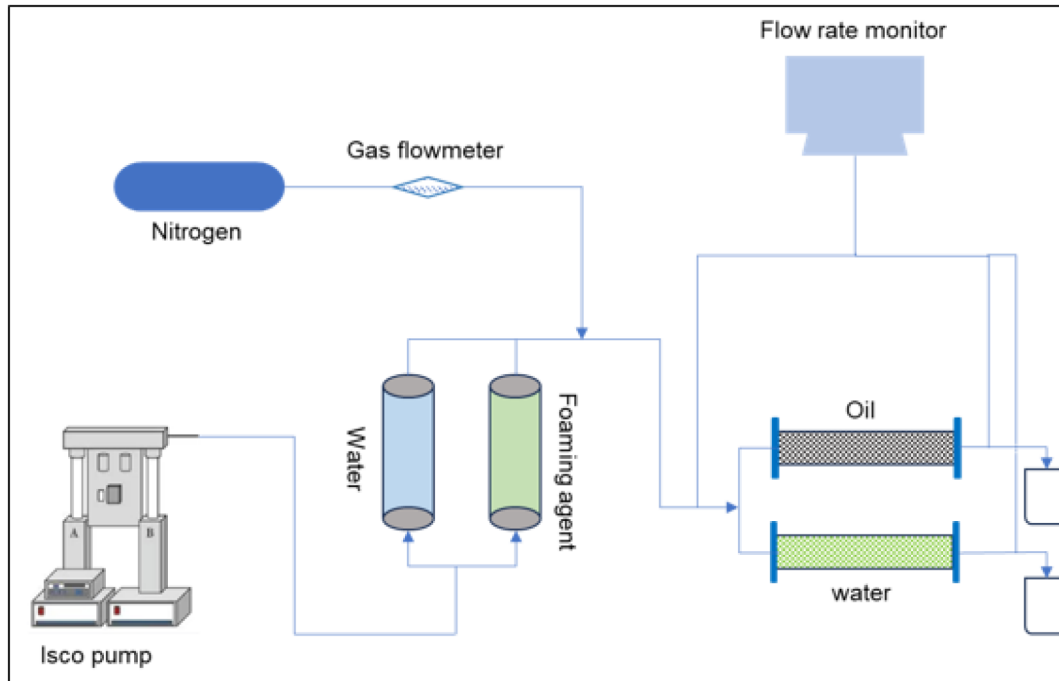


Figure 6—Parallel core flooding experiment to identify foam stability in water and oil core

In this experiment, at the beginning of injection, the liquid production rate from the water-saturated core was higher than that from the oil-saturated core. However, after approximately 1 PV of foam slug, the production profile shifted, and during the subsequent 7 PV of foam injection, the oil-saturated core consistently exhibited a higher liquid production rate than the water-saturated core, shown in Fig. – 7. Switching to water injection at about 8 PV of foam slug, the liquid production from the water-saturated core declined, with most of the liquid ultimately produced through the oil-saturated core. These results indicate that foam stability differs between the two environments: foam is unstable in the presence of oil and thus provides poor blocking effect in the oil-saturated core, while it remains stable in the water-saturated core, where sufficient foam injection can establish an effective blocking effect to inhibit water flow.

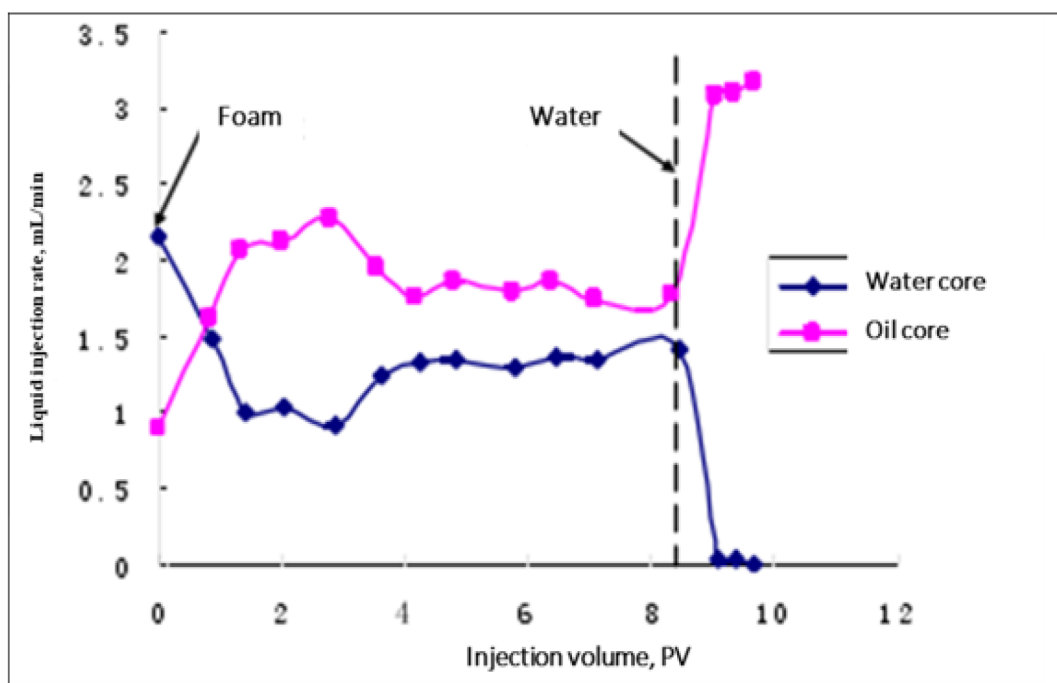


Figure 7—Liquid injection rate after the foam injection

Foaming Agent Selection

In this section, due to the high salinity of the formation water, several tests were carried out to evaluate and optimize the performance of the foaming agent. These included a compatibility test between the foaming agent and formation water, a foaming ability test, and optimization of the foaming agent concentration. All tests were conducted using formation water from the oilfield. The foaming ability and stability were assessed using the ‘Waring blender’ method. The results of the compatibility and foaming ability tests are summarized in Table 2.

Table 2—Foaming agent-screening result

Foaming agent	Phase behavior	Foam volume (ml)	Foam half-life (min)
DLY-05	Turbidity	/	/
HS-12	Turbidity	/	/
DXY-06	Clarified	380	11
DXY-08	Clarified	430	16

According to the test results, DXY-08 demonstrated the best overall performance and was therefore selected for further concentration optimization. In this experiment, seven different concentrations of DXY-08 were tested, and the results are presented in Fig. – 8. Based on the findings, the optimal concentration range was identified as 0.5% to 1%. For field application, a concentration between 0.5% and 0.7% is recommended, taking into account both technical effectiveness and economic feasibility.

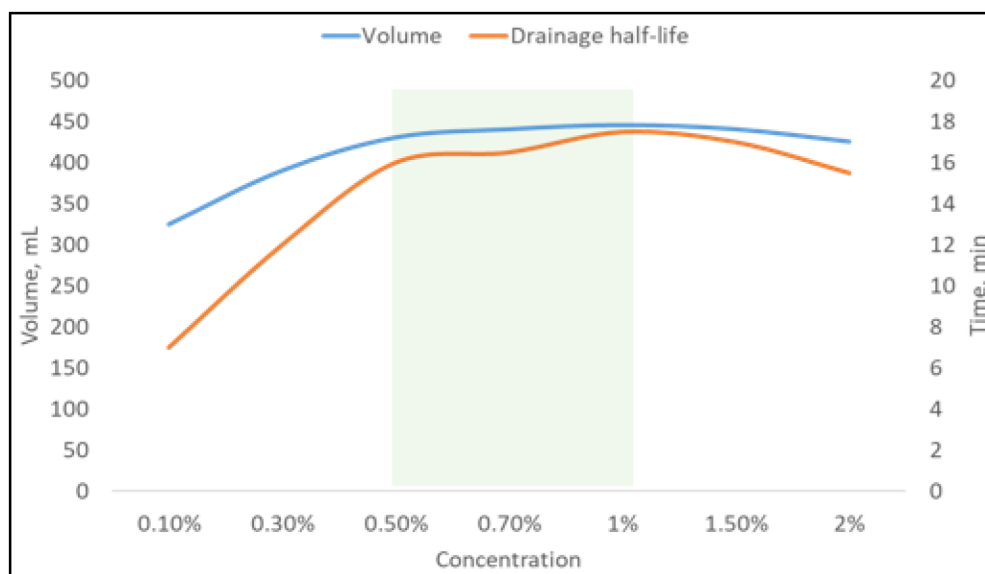


Figure 8—Foam volume and half-life time at different concentration

Gas–liquid Ratio Optimization

According to the experimental data, the foam exhibits the strongest plugging ability when the gas–liquid ratio is in the range of 1:1 to 3.5:1, with resistance factors between 120 and 130, shown in Fig. – 9. When the gas–liquid ratio exceeds 3.5:1, the resistance factor begins to decline, indicating reduced blocking performance. Therefore, the optimal gas–liquid ratio range is determined to be 1:1 to 3.5:1.

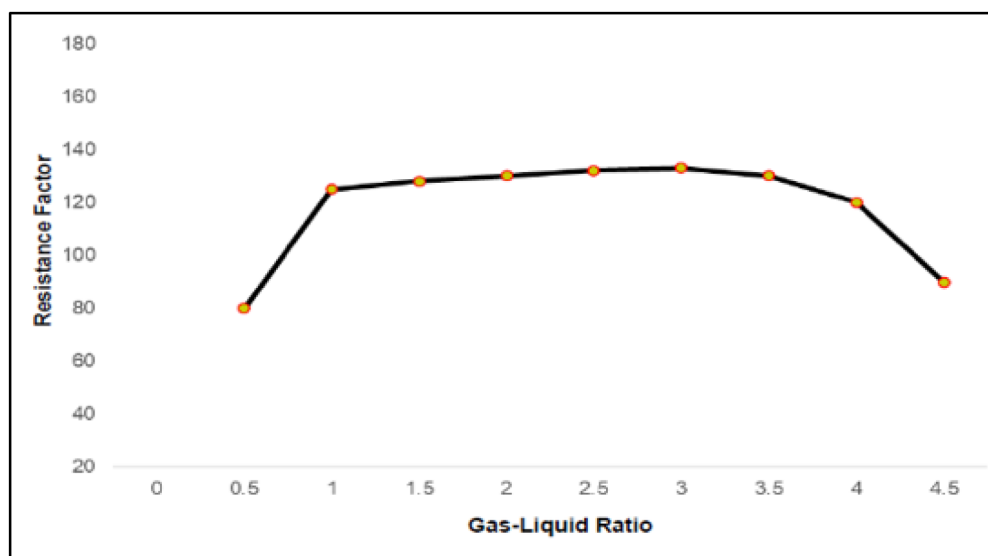


Figure 9—Gas to liquid ratio

Interfacial Tension Test

This test was conducted to determine whether the foaming agent could reduce the interfacial tension (IFT) between crude oil and formation water. The measurements were carried out using a TX-500C IFT instrument. IFT values were tested at six different concentrations: 0.05%, 0.1%, 0.15%, 0.2%, 0.25%, and 0.3%. The results showed that when the concentration reached 0.1% or higher, the IFT dropped below 0.1 mN/m. This means that, in addition to its foaming ability, the foaming agent also possesses interfacial tension reduction capabilities, which can enhance the oil displacement efficiency, shown in Fig. – 10.

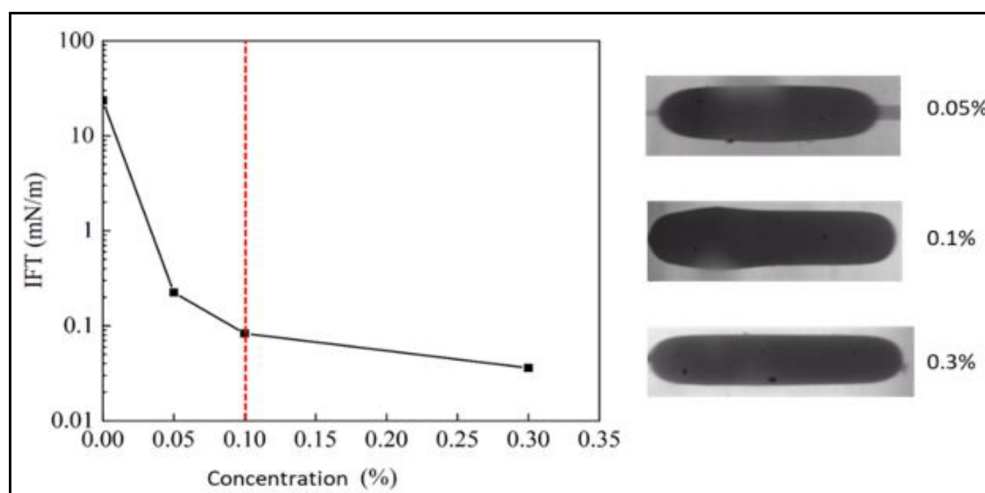


Figure 10—IFT value at different concentration

High Temperature and Pressure Impact

Foam, whose stability is highly influenced by temperature and pressure, is a thermodynamically unstable system. Typically, increasing temperature negatively affects foam stability, while increasing pressure has a positive effect. To investigate foam stability under high-temperature and high-pressure conditions, a supercritical PVT (Pressure-Volume-Temperature) instrument was used to conduct a series of crossover experiments in the laboratory, involving four temperature levels and three pressure levels. The results show that foam stability decreases with rising temperature and improves with increasing pressure. At a given pressure, there exists a critical temperature above which foam stability declines. This behavior is illustrated in Fig. – 11.

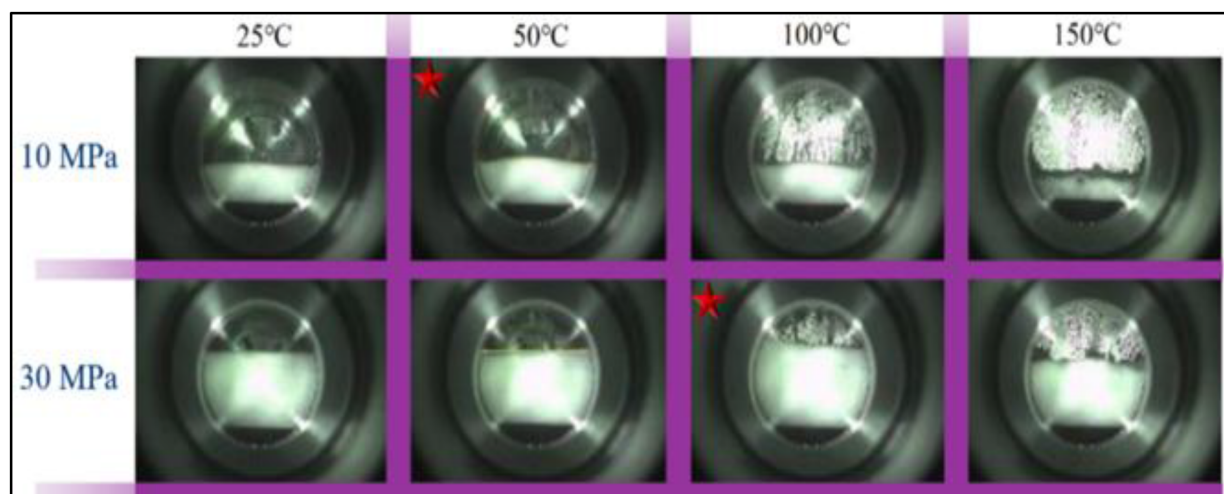


Figure 11—Foam stability at the condition of high temperature and high pressure

Laboratory evaluation provides critical evidence of foam performance and guides the selection of the optimal formulation for field application, and hence set the technical foundation for designing an injection proposal that can be practically implemented in the reservoir.

Injection Proposal

With lab validation completed, the next step was to translate the findings into a practical field-scale injection plan regarding injection pressure, volume, scheme, process and equipment.

Injection Pressure

The fluid and gas column pressure was simulated by PIPESIM, shown in below Fig. – 12. The simulated wellhead injection pressures range is from 1043.4 to 1696 Psi (7.4 to 11.7Mpa), which is below the reservoir fracturing pressure (3300 psi) during the foam injection.

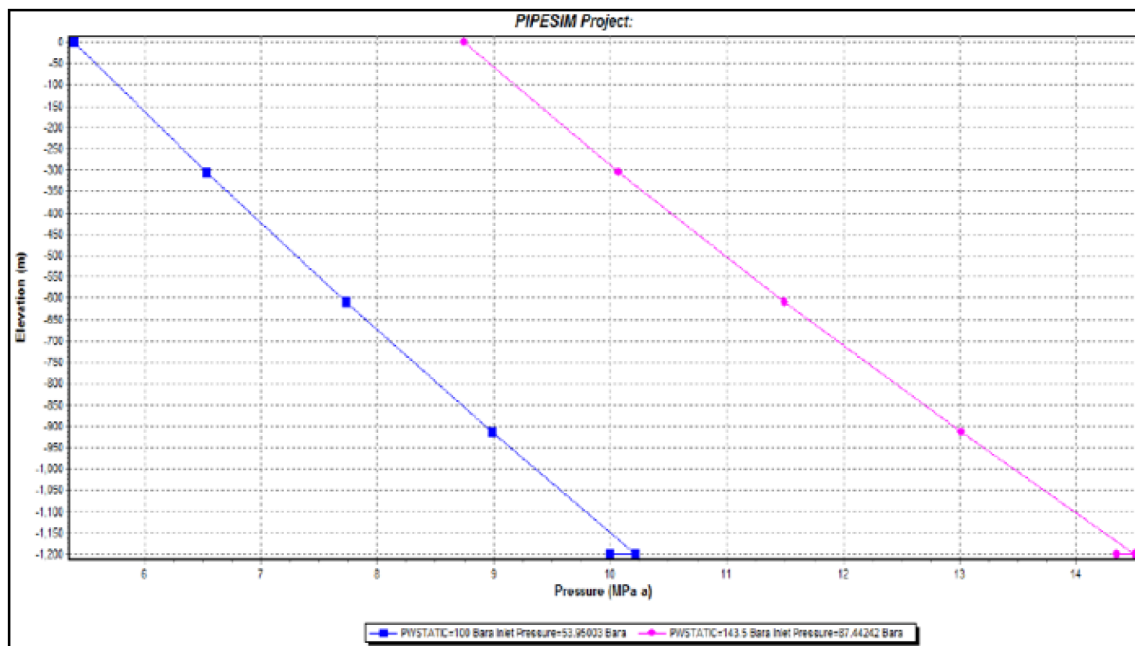


Figure 12—Injection pressure prediction result

Injection Volume

The nitrogen foam injection volume was calculated based on the injector's current monthly water injection rate. According to the field experience in other carbonate reservoir, the total injection volume of foam slugs is expected to be analytically similar with two-months water injection volume:

- In order to reduce the cost, minimum gas-liquid ration was used, and hence the selected gas-liquid is 1:1 according to the lab experiment.
- The historical monthly water injection volume was around 3,200m³. Therefore, the total volume of the N₂ foam in two months is 6,400m³.
- With a gas-liquid ratio of 1:1, the underground N₂ volume is around 3,200m³, which means the N₂ volume is around 355,000 m³ (11.5 MMscf) at surface.
- With a gas-liquid ratio of 1:1, the total volume of the foam surfactant solution should be 3,200m³.

Injection Scheme

- Day 1: Stop the water injection and inject 2,500 m³ of N₂ gas to pressurize the reservoir. Meanwhile, it can test the gas injectivity and injection pressure. Inject 6,500 m³ of N₂ gas along with 50 m³ of foam Surfactant at 1% concentration simultaneously to create the foam.
- Day 2-25: Repeat the Simultaneous injection of gas at 14,420m³/d and foam Surfactant solution at 135 m³/d with 0.7% concentration for 24 days.
- Inject 200 m³ of water to flush the reservoir and then resume the water injection.

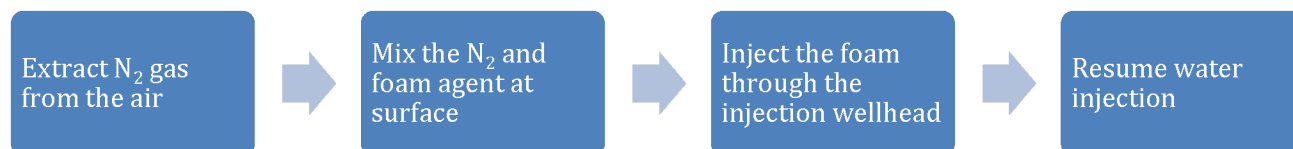
Actual field trial numbers are shown in Table 3.

Table 3—Foam injection proposal

Slugs	Injection process	Injection Amount	Proposed Chemical Concentration	QTY of Chemical powder
Pre-Gas slug	Continuous Injection of Pure N ₂	2,500 m ³		
Pre-Foam slug	Continuous Injection of N ₂	6,500 m ³	-	-
	Foam solution	50m ³	1%	0.5 tons
Foam slug	Continuous Injection of N ₂ at 14,420 Nm ³ /Day	346,000 m ³	-	-
	Continuous Injection of foam solution at 135m ³ /Day	3,240 m ³	0.70%	19.5 tons
Displacement slug	Chase water	200 m ³	-	-

Injection Process

Foam generation was planned at the surface, and then injected into the reservoir through the injection wellhead according to below injection process:



Nitrogen Gas Extraction from Air. Nitrogen (N₂) is extracted from atmospheric air using a nitrogen generation unit, such as a membrane separation system. This system separates nitrogen from oxygen and other trace gases to produce nitrogen with high-purity of more than 98%.

Surface Mixing of Nitrogen and Foaming Agent Solution. The extracted nitrogen gas is then mixed with an aqueous foaming agent solution at the surface. This foaming agent typically consists of surfactants specifically formulated to generate and stabilize foam under reservoir conditions. The mixing process is carefully controlled to produce a stable nitrogen foam with the gas–liquid ratio of 1:1.

Foam Injection into the Reservoir. The generated nitrogen foam is injected into the reservoir through the wellhead of designated injector. The injection is performed over 25 days, as outlined in the injection plan. The goal is to improve sweep efficiency by selectively blocking high-permeability channels and diverting subsequent fluids into lower-permeability zones.

Transition to Water Injection. After the planned foam injection, the injection operation is switched back to conventional water flooding. By this stage, the foam is expected to have established a strong blocking effect in high-permeability zones, thereby enhancing the overall oil displacement efficiency during the subsequent water injection phase.

Injection Equipment

Several injection facilities (shown in Fig. – 14), such as aircompressor, nitrogenmembrane separation unit, nitrogen booster, foam generator and injection pumps with the technical specifications in Table 4, are placed on the well site to conduct the foam injection as per the schematic diagram (shown as Fig. – 13).

Table 4—The specification of the injection

Equipment name	Technical specification	Quantity
N ₂ equipment	MZD-1200/350Q	1
Foam generator		1
Water injection pump	8-10 m3/hr	1
Liquid mixing tank	20 m3	2
Duty room	5.9m*2.4*2.4	1

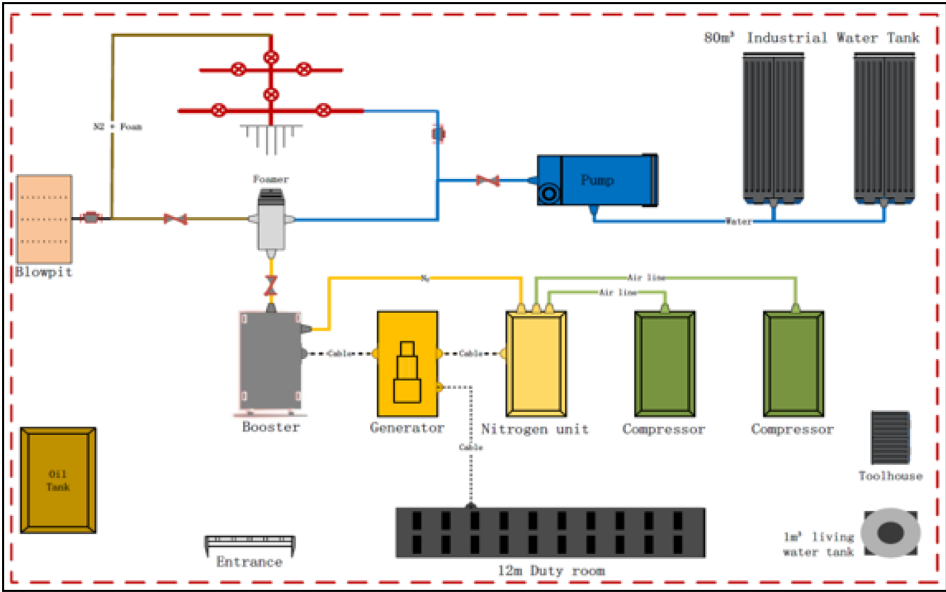


Figure 13—Schematic diagram of the injection process



Figure 14—The injection facility

Pattern Summary

Designed nitrogen foam systems can be effective in medium–high permeability reservoirs (Liu, Y., et al., 2023); but for tight carbonate reservoir, the challenge in field piloting is high. A well pattern was selected for the pilot trial including one main injector WI1 and two key offset producers (PROD1 and PROD2) at 50m spacing as shown in Fig. – 15. The injector commenced water injection in 2014 and had cumulatively injected over 1.5 million barrels of water prior to the foam trial. Offset wells PROD1 and PROD2 showed declining oil rates and rising water cuts prior to intervention, with PROD1 averaging gross of 195 bpd at 81% water cut, and PROD2 at gross of 247 bpd with 76% water cut. The geological configuration between these wells indicated a high-permeability zone that may have caused preferential flow paths during water flooding, leading to early water breakthrough and bypassed oil. The wells in this pattern were drilled with a closer spacing of 50 meters, unlike many others in the area, resulting in a significantly stronger impact of injection water on the oil producers.

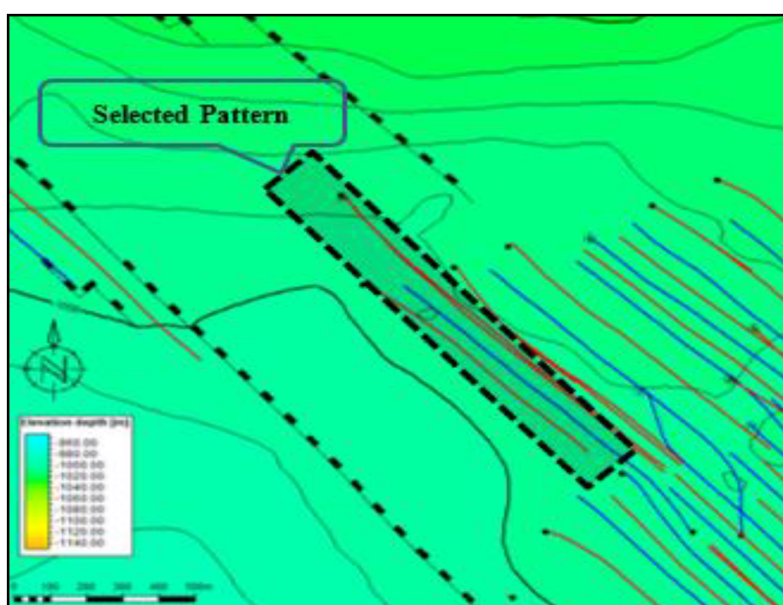


Figure 15—Selected Pattern

The selected pattern showed clear injector-producer relationships, low cumulative oil recovery, and moderate daily fluid rates, making it an ideal candidate for a foam flooding trial. Before initiating the foam trial, a comprehensive pre-job evaluation was conducted to assess the feasibility and potential benefits of foam injection. This evaluation process included the following critical steps:

1. **Historical Production Analysis:** A detailed review of historical production data for PROD1 and PROD2, as well as the performance trends of the WI1, was carried out. This analysis helped to establish a baseline understanding of the well behavior prior to the intervention, focusing on the water cut progression and declining oil rates, both of which suggested a need for intervention.
2. **Surveillance Data Review:** A thorough examination of the surveillance data, including pressure and injection rate data from the WI1 well and offset producers, was essential. This data helped identify pressure anomalies and discrepancies in water injection distribution, which may have been contributing to the observed early water breakthrough.
3. **Laboratory Foam Compatibility Testing:** To ensure that foam flooding could be an effective solution for the reservoir, foam compatibility tests were conducted in the laboratory using Daleel reservoir fluids. These tests assessed how well foam would perform with the specific fluid properties present in the reservoir, including oil viscosity, water salinity, and reservoir pressure conditions.

The combination of geological understanding, production performance, and foam compatibility testing led to the selection of the selected pattern for the foam flooding trial, as it showed the most potential for benefiting from this EOR method.

Field Trial Performance

Once the injection proposal was executed, both injection and production responses were carefully monitored to evaluate the effectiveness of nitrogen foam flooding.

Injection Performance

The nitrogen foam injection commenced in September 2024 and was executed in three phases:

Pre-Slug Phase: Pure nitrogen gas (2,500 m³) was injected to compensate for formation pressure and establish baseline for the gas injectivity and pressure within the reservoir for 4 hrs. Then inject 6,500m³ of N₂ gas along with 50 m³ of foam surfactant at 1% concentration simultaneously for 7 hrs.

Foam Slug Phase: Over 24 days, nitrogen gas and foaming solution were co-injected continuously. The total nitrogen volume was 346,000 m³, while 3,240 m³ of foaming agent solution was injected at the concentration of 0.5% to 0.7%. The gas-liquid ratio was maintained between 1:1 and 1.2:1.

Fig. – 16 and Fig. – 17 show the proposed and actual foam concentration, and the proposed and actual nitrogen's gas injection rate, respectively.

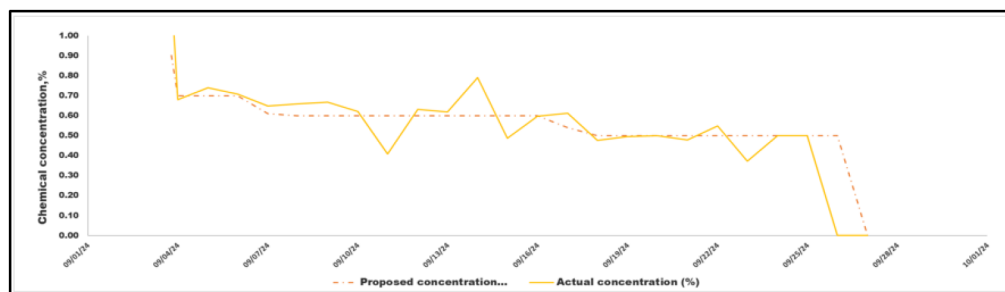


Figure 16—Foaming agent concentration



Figure 17—N₂ gas injection rate

Displacement Slug: Following foam injection, 200 m³ of chase water was injected to push the foam deeper into the formation and enhance the contact with residual oil zones.

At one-month foam injection stage, as shown in Fig. – 18, injection wellhead pressure reached up to 13.5 MPa (2,000 psi), and then the pressure buildup of the injector had been maintained for around 2-3 months. Meanwhile, no water breakthrough occurred in any of the producers.

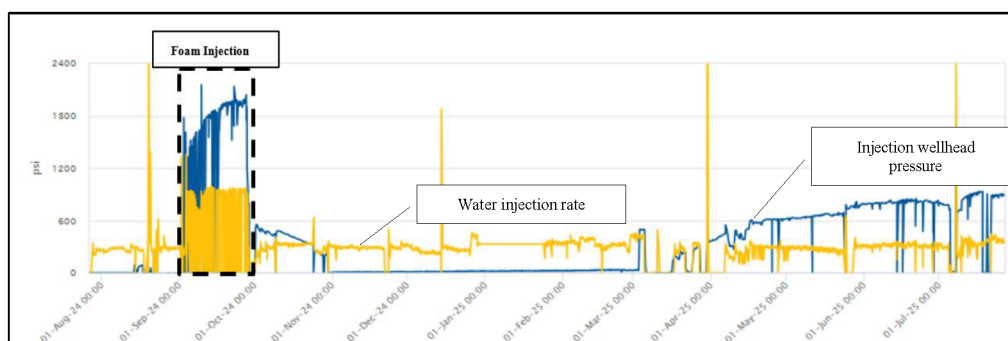


Figure 18—Water Injection wellhead pressure and injection rate

Production Performance

The nitrogen foam flooding trial in the selected pattern provided encouraging results, validating its potential as an EOR method in mature carbonate water flood. The pilot delivered tangible production gains, improved water management, and demonstrated foam's capacity to influence reservoir conformance. Post-injection monitoring showed a positive response in both producers, though the behavior varied:

PROD1: This well exhibited a stable and sustained oil production increase, with cumulative incremental oil reaching 3,889 bbls by June 2025. The water cut declined and then stabilized, suggesting successful foam diversion and improved sweep, Fig. – 19.

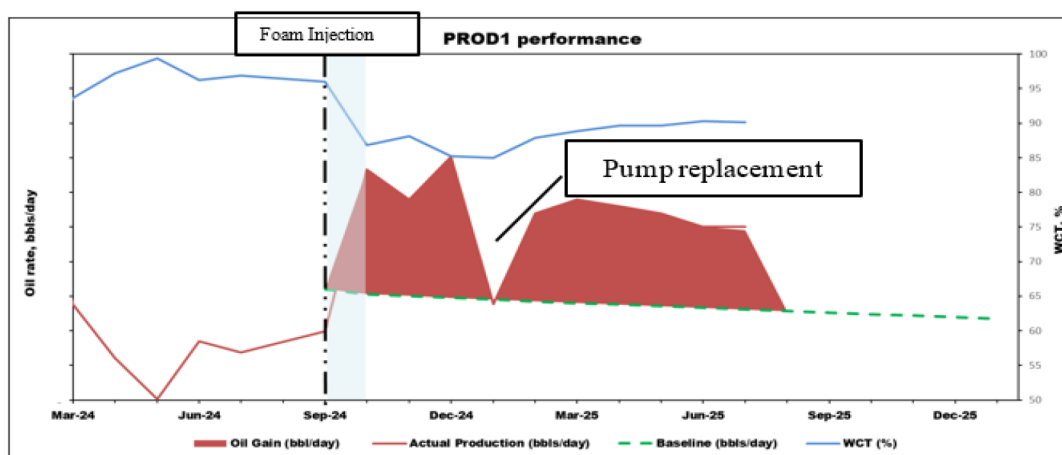


Figure 19—Production Performance of PROD1

PROD2: Although oil production increased, the performance was more erratic. Incremental oil reached 2,741 bbls by June 2025, but water cut began rising again after 6 months, Fig. – 20. This may be due to limited foam propagation or a more complex permeability structure.

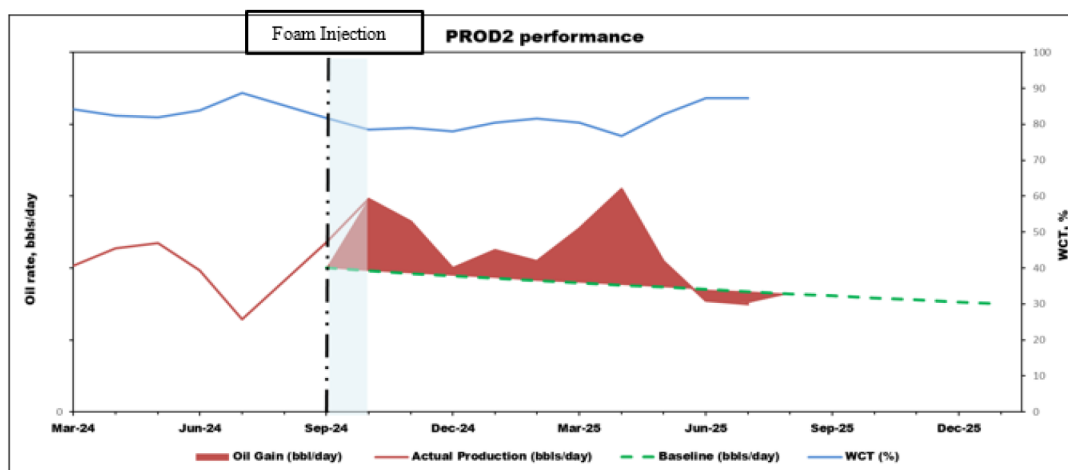


Figure 20—Production Performance of PROD2

Overall, the pilot achieved 6,630 bbls of incremental oil over a 9-month period. The water cut reduction ranged between 8% and 15%, with the effect reaching the peak within 3–4 months.

Discussion of the Results

Below observations revealed key insights into foam behavior and reservoir response, which not only explain the performance trends of the pilot but also highlight practical considerations for optimizing future foam flooding projects.

Quick Pressure Response

Upon the start of the foam injection, a quick increase in the pump intake pressure of both producers were seen in 5 days, shown in Fig. – 21, which means the foam slug can replenish the reservoir pressure rapidly.

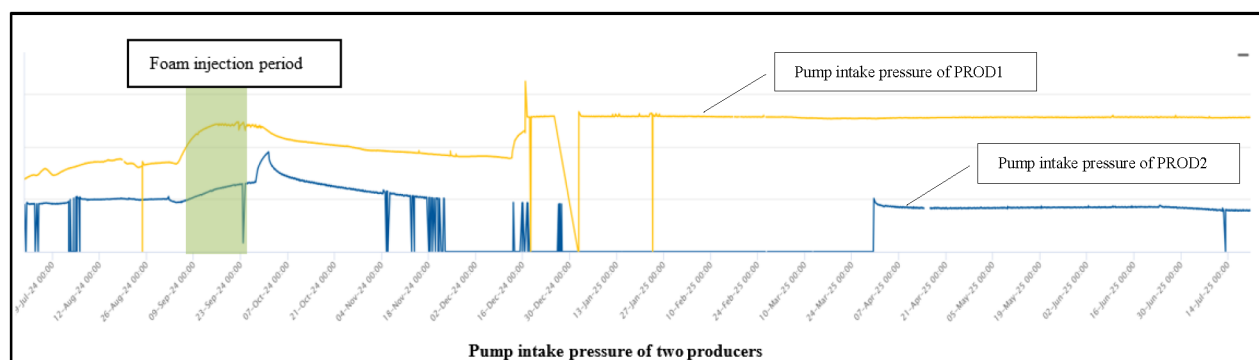


Figure 21—Pump intake pressure of both producers

Gradual Gas Release

Below, Fig. 22, shows a fluctuation in the pump fill-age curve in PROD1 for almost 3 months, which means that there is a gradual gas release up to December 2024. During the three months, the injection wellhead pressure of the injector and pump intake pressure of the producer declines gradually as well. In order to control the gas release and maintain a longer pressure buildup, it is recommended to add more viscous gel/polymer to improve the foam stability.

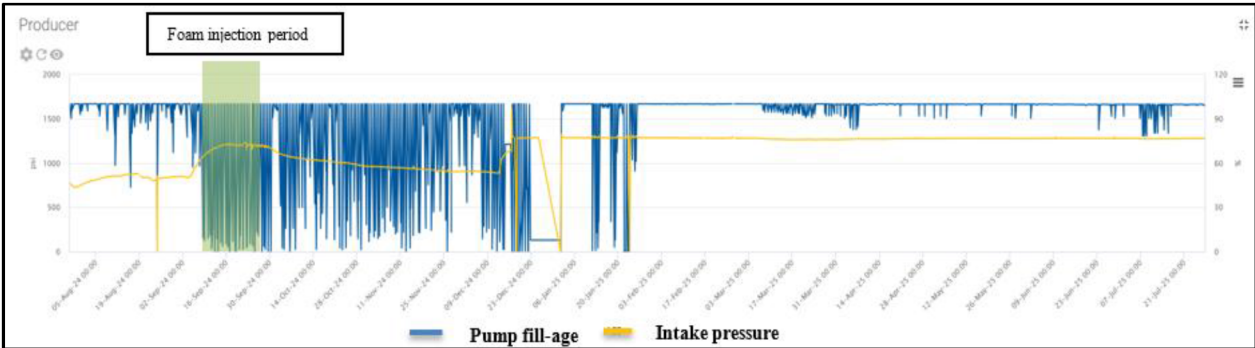


Figure 22—Pump fill-age curve and intake pressure of PROD1

Nearby Well Interference

At the injection stage, clear pump intake pressure response is seen in a nearby producer located 50m outside the pattern, meaning that gas has very good mobility to transfer faster and broader to cause unexpected interference in the nearby producers. To avoid the uncertainty brought by the producers out of the pattern, it is highly recommended to select a clean and proper well pattern where the communication between the producers and injectors are very clear.

Memory Production Logging Test Result

Based on the 2025 log results and a comparative analysis of high-precision temperature and injection profiles before and after the Foam injection, the data indicates a redistribution of injection. Zones 1 and 2 (1417.0-1613.5 m) exhibit reduced cooling effect compared to the 2024 log (pre-injection), while zones 3 and 4 (1613.5-2179.0 m) demonstrate cooling increase (refer to Fig. – 23). Thus, the redistribution of the injection towards to the second half of the open hole section should be considered.

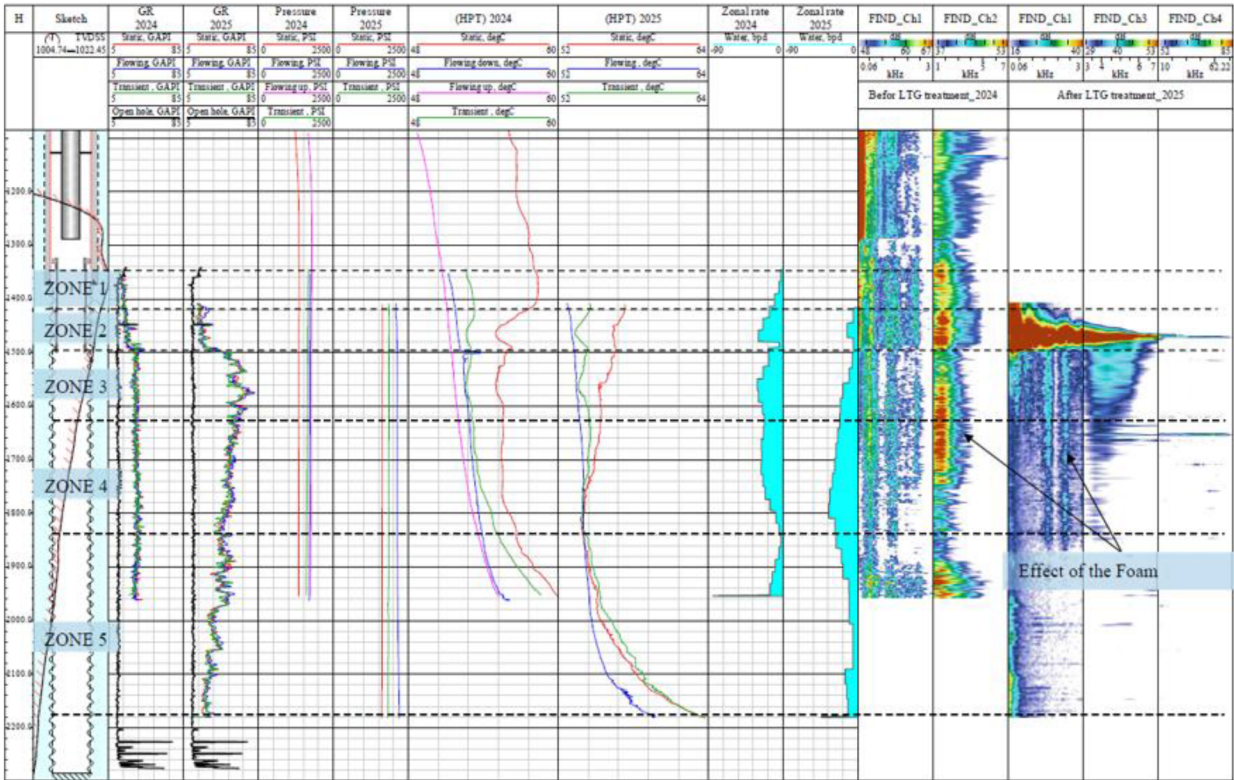


Figure 23—Memory production logging test result

It is observed that, according to the MPLT at zone 2, less water injection was lost behind the short liner into possible fault indicating plugging of the thief zone by the foam.

Conclusions

The findings from this pilot trial consolidate the role of nitrogen foam as a promising EOR technology for tight carbonate reservoirs. In summary, nitrogen foam flooding has proven technically feasible and economically attractive in the Natih reservoir, providing a foundation for scale-up and continued optimization in future field applications.

- Both single-core and dual-core parallel flooding experiments were conducted to evaluate the blocking ability of nitrogen foam, indicating that the foam provided stronger blocking ability in the high-permeability zone than the low-permeability zone.
- Foam is unstable in the presence of oil and thus provides poor blocking effect in the oil-saturated core, while it remains stable in the water-saturated core, where sufficient foam injection can establish an effective blocking effect to inhibit water flow.
- In addition to its foaming ability, the foaming agent also possesses interfacial tension reduction capabilities to the level of 0.1 mN/m, which can enhance the oil displacement efficiency as well.
- Foam, whose stability is highly influenced by temperature and pressure, is a thermodynamically unstable system. Typically, increasing temperature negatively affects foam stability, while increasing pressure has a positive effect.
- A clear increase in the pump intake pressure and injection wellhead pressure was seen in 3 months upon the foam injection, which shows the obvious pressure buildup in the reservoir.
- Memory production logging test (MPLT) was performed before and after the foam injection. The result illustrates a redistribution of the water injection based on the log results form a comparative analysis of high-precision temperature and injection profiles.
- A clear water cut drop was found in both injectors, lasting for around 4 months. In total, 6,630 barrels of incremental oil were produced, providing an encouraging result that supports the implementation of a second field trial with additional injection wells.
- Further improvements in the foam slugs and chemical formula, particularly to enhance foam stability over longer durations, are necessary for any future follow-up operations.

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